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Forensic Artefact Analysis of Cryptocurrency Wallet
Applications on Android: Recovery, Persistence, and
Evidential Value

Faculty of Engineering and Applied Sciences
Digital Forensics

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requirements for the degree of MSc.

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- the thesis submitted has not been previously submitted to this university or any other.
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Abstract

Cryptocurrency wallet applications on Android devices are an increasingly common source of digital evidence, linking a device, its user, and blockchain-recorded activity. Prior research has catalogued individual artefacts recoverable from wallet apps – wallet identifiers, transaction records, timestamps, and account metadata – but has given comparatively little attention to whether such artefacts persist after user actions such as logout, deletion, cache clearing, uninstallation, or backup and restore, and to how strongly they support investigative conclusions. This thesis addresses that gap through a controlled, phased forensic examination of three Android cryptocurrency wallet applications spanning EVM-based, multi-chain, and Bitcoin-focused wallet designs. Each application is taken through a fixed sequence of user actions – installation, wallet creation and import, transaction activity, synchronisation, logout, deletion, cache and storage clearing, and uninstall/reinstall – with forensic acquisition performed at each stage. Recovered artefacts are classified by persistence state and assessed against a structured evidential value framework distinguishing artefacts that support association between device, application, and wallet address from those that would additionally require corroboration to support a claim of ownership or control. The intended contributions are a reproducible artefact recovery and persistence framework for Android cryptocurrency wallets, a comparative acquisition analysis, and practical guidance for investigators on evidential weight and for developers on secure-by-default storage behaviour.

Keywords:

Digital forensics; Android forensics; cryptocurrency wallets; mobile forensics; artefact persistence; evidential value; blockchain evidence; data recovery

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List of Abbreviations

FEAS	Faculty of Engineering and Applied Sciences
ADB	Android Debug Bridge
APK	Android Package (application file format)
BIP	Bitcoin Improvement Proposal
CURES	Cranfield University Research Ethics System
EVM	Ethereum Virtual Machine
FBE	File-Based Encryption
NIST	National Institute of Standards and Technology
TEE	Trusted Execution Environment
UTXO	Unspent Transaction Output
WAL	Write-Ahead Log (SQLite)

Chapter 1

Introduction

1.1 Background and Motivation

Cryptocurrency wallet applications installed on Android devices sit at the intersection of mobile forensics and blockchain-based evidence. A wallet application can connect a physical device to a user's activity, to one or more blockchain addresses, to counterparties, and to a timeline of transactions – making it a potentially rich evidence source in investigations ranging from fraud and money laundering to ransomware payment tracing. Mobile forensic practice is well established for structuring this kind of investigation, framing work around preservation, acquisition, examination, analysis, validation, and reporting under forensically sound conditions **[CITE: NIST SP 800-101 Guidelines on Mobile Device Forensics – add to MyReferences.bib once verified]**.

Prior work examining Android cryptocurrency wallets has demonstrated that recoverable artefacts can include wallet identifiers, transaction identifiers, timestamps, associated email addresses, cookies, and OAuth tokens. However, this body of work has tended to catalogue what can be found on a device at a single point in time, with comparatively little systematic attention paid to two questions that matter directly to investigative practice: whether an artefact survives ordinary and deliberate user actions taken after the fact, and how much evidential weight an examiner is entitled to place on it once found.

1.2 Research Gap

Existing studies have identified artefacts within selected cryptocurrency wallet applications, but there is limited comparative research on modern Android wallet apps that examines recovery state, persistence following user actions, and evidential value together, across more than one application design (EVM-based, multi-chain, and UTXO-based wallets differ structurally in ways that plausibly affect what is recoverable). Many studies list artefacts found under a single acquisition condition; fewer assess whether those artefacts survive deletion,

logout, cache clearing, uninstallation, or backup and restore – and fewer still translate a recovered artefact into a defensible statement about what it does and does not prove.

1.3 Aim

To identify, recover, classify, and evaluate forensic artefacts produced by selected Android cryptocurrency wallet applications, with particular focus on artefact persistence after user actions and their evidential value for wallet attribution, transaction reconstruction, and investigative timelines.

1.4 Research Questions

- RQ1** What forensic artefacts are generated by selected Android cryptocurrency wallet applications during installation, wallet creation/import, transaction activity, synchronisation, logout, deletion, and uninstall?
- RQ2** Which artefacts persist after user actions such as clearing cache, deleting wallet/account, clearing app storage, uninstalling/reinstalling, and backup/restore?
- RQ3** How do different acquisition approaches affect recoverability of cryptocurrency wallet artefacts on Android?
- RQ4** What evidential value do recovered artefacts provide for attribution, timeline reconstruction, transaction verification, and investigative reporting?

1.5 Scope and Boundaries

This thesis does not attempt to recover private keys or seed phrases from production wallets, compromise real wallets, or provide a general-purpose key-extraction method. All experimentation uses researcher-controlled test wallets and testnet transactions on an Android emulator. Where a commercial forensic suite is not available through the university, RQ3 is scoped to compare logical/ADB-level acquisition against rooted full-filesystem acquisition rather than assuming access to tools such as Cellebrite or Magnet AXIOM; this boundary is confirmed in Chapter 3.

1.6 Thesis Structure

Chapter 2 reviews Android forensics fundamentals, cryptocurrency wallet architectures, Android storage and security mechanisms (file-based encryption, Android Keystore), prior wallet-forensics studies, and the evidential-value

literature. Chapter 3 sets out the experimental design, wallet and device selection, the phased action sequence, acquisition strategy, artefact taxonomy, evidential value framework, and ethical considerations. Later chapters (not yet drafted, pending experimental work) will cover the experiment setup in full technical detail, present results per application, analyse persistence, discuss evidential value, and conclude by answering each research question in turn.

Chapter 2

Literature Review

[Structural skeleton only – populate with real sources as reading progresses. No citations below are real; do not cite this chapter until sources are added to `MyReferences.bib`.]

2.1 Mobile Forensics Fundamentals

Cover: standard mobile forensic process models (preservation/acquisition/examination/analysis/validation/reporting); acquisition tiers (logical, file-system, physical) and where each sits relative to modern Android security; chain of custody and forensic soundness expectations for taught-MSc-level work.

2.2 Android Storage and Security Architecture

Cover: app-specific internal storage, shared preferences, SQLite (including WAL/SHM behaviour), cache vs. files vs. `no_backup` directories; Android Auto Backup inclusion/exclusion rules; File-Based Encryption and the Credential Encrypted vs. Device Encrypted storage distinction; Android Keystore and hardware-backed key protection (TEE/Secure Element).

2.3 Cryptocurrency Wallets: Architecture and Key Management

Cover: custodial vs. non-custodial wallet models; BIP-39 mnemonic/seed generation; EVM account model vs. Bitcoin's UTXO model; how wallet apps typically structure local state (per-app, so note this must be confirmed empirically rather than assumed from architecture diagrams alone).

2.4 Prior Forensic Studies of Cryptocurrency Wallets

Cover: existing artefact-recovery studies on Android/desktop/browser-extension wallets; what has and has not been tested for persistence; gaps this thesis targets.

2.5 Blockchain Evidence and Attribution

Cover: blockchain analytics and address-clustering literature; known limitations and risks of attribution assumptions; how public on-chain data relates to (and is distinct from) device-side evidence.

2.6 Evidential Value and Digital Evidence Reliability

Cover: frameworks for weighting digital evidence; the association-vs-control distinction central to this thesis's RQ4; standards for expressing forensic conclusions defensibly (relevant to the viva defence of methodology).

2.7 Summary and Positioning

Close by restating the research gap from Section 1.2 in light of the literature actually reviewed, and confirm RQ1–RQ4 remain the right questions once the literature is in.

Chapter 3

Methodology

3.1 Research Design

This thesis uses a controlled experimental design: each selected wallet application is taken through an identical, fixed sequence of user actions on a dedicated test device, with a forensic acquisition performed at each stage. This produces a phase-by-phase record of artefact creation and, critically, artefact survival, which subsequent phases can then be checked against. The design is repeatable across applications, which supports the comparative aim in RQ1–RQ3, and produces the evidence base needed to answer RQ4.

3.2 Wallet Application Selection

Three non-custodial wallet applications are selected, one from each of the following categories, chosen so that the comparison spans materially different account and storage models rather than three variations on the same design:

- An EVM/Web3 wallet (candidate: MetaMask Mobile), representing the Ethereum account model and EVM-compatible chains.
- A multi-chain wallet (candidate: Trust Wallet or Coinbase Wallet), representing apps that manage several distinct chain architectures within one application.
- A Bitcoin-focused wallet (candidate: BlueWallet or Bitcoin Wallet), representing the UTXO transaction model, which differs structurally from account-based chains.

A fourth application, such as a custodial exchange app, is treated as a stretch goal only, given the additional KYC and cloud-account complexity such apps introduce and the limited time available. Non-custodial wallets are preferred throughout because their artefacts relate directly to on-device wallet creation,

key handling, and locally stored blockchain data, rather than to server-side account records.

Each candidate application’s actual local-storage implementation is confirmed empirically during the pilot phase (Section 3.5) rather than assumed from documentation, since app architecture (for example, React Native vs. WebView-based apps) affects where and how artefacts are stored on Android.

3.3 Device and Environment

Testing uses an Android Studio emulator (Android Virtual Device) rather than a physical handset, which allows use of a Google APIs/root-enabled system image for full-filesystem access to `/data/data/<package>` where required, while keeping the environment fully reproducible and disposable between test runs. The specific Android API level, device profile, and system image used are recorded in the experiment log so results can be reproduced exactly.

3.4 Phased Experimental Procedure

Each wallet application is carried through the following fixed action sequence, with a forensic acquisition performed at each labelled point:

Table 3.1: Phased action and acquisition sequence

Phase	User action	Acquisition point
Baseline	Fresh emulator image, no wallet installed	A0
Install	Install wallet app, no account created	A1
Create wallet	Create new test wallet, set PIN/biometric	A2
Import wallet	Import test mnemonic into app	A3
Receive transaction	Receive testnet crypto	A4
Send transaction	Send testnet transaction	A5
Add token/network	Add custom token/network/dApp connection	A6
Lock/logout	Log out, lock wallet, close app	A7
Delete wallet	Use in-app delete/remove wallet option	A8
Clear cache	Android settings → clear cache	A9
Clear storage	Android settings → clear storage	A10
Uninstall/reinstall	Remove app and reinstall	A11
Backup/restore	Test Android backup/restore behaviour	A12

Android Auto Backup, by default, can include shared preferences, internal files, databases, and external app-specific files, while cache, code cache, and `no_backup` directories are excluded; individual apps may also disable backup entirely. Non-custodial wallet apps routinely do so, given they handle key material, so phase A12 is expected to return a “not testable” or “no artefacts

transferred” result for some or all of the tested applications. This is treated as an expected, security-relevant finding rather than an experimental failure, and each application’s backup manifest flag is checked and recorded ahead of the full run.

3.5 Pilot Phase

One wallet application is carried through the full phase sequence first as a pilot, to surface and resolve procedural issues – confirmation of storage backend, testnet transaction reliability, timing of synchronisation, and acquisition tooling – before the remaining applications are run. The method is adjusted once, after the pilot, and then held fixed for the remaining applications to preserve comparability.

3.6 Acquisition Strategy

At least two acquisition tiers are used at each phase:

- **Logical/visible acquisition** – UI screenshots and app-exported data, representing what a basic examination would see.
- **File-system acquisition via ADB** on the rooted emulator image, reaching app-specific storage under `/data/data/<package>` where the Android security model permits.
- **Commercial forensic tooling**, where confirmed available through the university, used as a comparison point rather than the primary acquisition method.

Modern Android security is expected to constrain what is recoverable: Android 10+ mandates file-based encryption, Credential Encrypted storage is only accessible after device unlock, and Android Keystore is designed to keep key material non-exportable and, where hardware-backed, bound to a Trusted Execution Environment or Secure Element. Accordingly, the working expectation is that private keys and seed phrases will not be directly recoverable through the tested methods, but that wallet addresses, transaction identifiers, chain identifiers, dApp links, timestamps, cached balances, session identifiers, and configuration data will still be recoverable and evidentially useful – a result that would itself answer RQ1–RQ3 meaningfully even where it is a negative result for key material specifically.

3.7 Artefact Taxonomy and Classification

Artefacts are catalogued against a fixed set of storage locations (shared preferences; SQLite databases and their WAL/SHM files; LevelDB/IndexedDB or

framework-specific local-storage backends; WebView storage; cache; files/no_backup directories; Android Keystore references; notifications; clipboard; external storage; package metadata), and each recovered artefact is classified using the following persistence states.

Table 3.2: Artefact persistence classification

Status	Meaning
Active	Visible/recoverable after normal use
Persistent	Survives logout/delete/cache clearing
Deleted but recoverable	Found in WAL, cache, backup, or residual files
Metadata only	Address/timestamp exists but not full detail
Not recoverable	No artefact found by the tested methods
Requires elevated acquisition	Only via full filesystem/rooted extraction
Cloud-only	Not locally recoverable; needs service-side records

3.8 Evidential Value Framework

Not every recovered artefact carries the same evidential weight, and this thesis is deliberately built around that distinction rather than treating recovery as an end in itself. A transaction identifier stored on a device is useful, but does not by itself establish that the device's user controlled the wallet in question; blockchain analytics research cautions that attribution can be weakened by clustering heuristics and interpretation assumptions. Each artefact type is therefore scored against the scheme in Tab. 3.3, and findings are written using an association/control distinction throughout: a recovered artefact is reported as supporting an association between the examined device, the wallet application, and an observed address, with any claim of ownership or control flagged as requiring corroboration (transaction-signing activity, exchange records, communications, or admissions).

3.9 Tools

Android Studio Emulator; Android Debug Bridge (ADB); Autopsy and ALEAPP for artefact parsing; DB Browser for SQLite and `sqlite3` for database and WAL/SHM analysis; JADX, APKTool, and MobSF for app inspection; `ripgrep`/`strings`/CyberChef for text and binary search; Plaso/`log2timeline` for timeline construction where useful; `sha256sum`/`hashdeep` for hashing and integrity verification; public testnet block explorers for transaction validation; and a written chain-of-custody log and experiment diary maintained throughout.

Table 3.3: Evidential value scoring scheme

Value	Example	Interpretation
High	Address + transaction ID + timestamp + app record + on-chain confirmation	Strong support for app activity
Medium	Address in cache/preferences only	Association, not necessarily control
Medium	dApp URL + connected address	Supports interaction with service
Low	Token logo/cache file only	Contextual, weak alone
Low	Public blockchain transaction alone	Needs device/app/user corroboration
High but sensitive	Seed phrase screenshot (test device)	Strong but must be redacted

3.10 Ethical Considerations

Only researcher-owned test devices, researcher-created test wallets, and testnet funds are used; no real victim data, stolen wallet data, or third-party wallets are examined. Live seed phrases, private keys, authentication tokens, or reusable recovery procedures are not published, and sensitive values appearing incidentally in screenshots are redacted. The experimental dataset is stored securely for the duration of the project. Ethical approval for this protocol has been submitted via CURES; per the module's mandatory requirement, data collection does not proceed, and this thesis cannot be marked, without confirmed approval on file (see Appendix A).

3.11 Limitations

The use of an emulator rather than physical hardware is a deliberate reproducibility choice, but it may not perfectly replicate hardware-backed key storage behaviour on a production device, and this is treated as a stated limitation rather than a hidden one. Acquisition comparison under RQ3 is bounded by whatever tooling the university confirms as available; where a commercial suite is not accessible, RQ3 findings are scoped to logical vs. rooted file-system acquisition only. Findings are further limited to the specific application versions tested, which are recorded for reproducibility, since wallet apps update frequently and storage behaviour may change between versions.

Chapter 4

Experiment Setup

[To be completed following device/app finalisation – device specs, Android/API version, exact wallet app versions, test wallet and testnet transaction details, full experiment log.]

Chapter 5

Results

[To be completed following the phased experimental run – artefact tables per application, file paths, timestamps, database records, screenshots, hashes.]

Chapter 6

Persistence Analysis

[To be completed – what survived logout, deletion, cache clear, storage clear, uninstall, reinstall, and backup/restore, per application.]

Chapter 7

Evidential Value Discussion

[To be completed – attribution, reliability, limitations, false positives, and corroboration needs, applying Section 3.8 to the actual results.]

Chapter 8

Conclusion and Future Work

[To be completed – summary of contributions, an explicit answer to each of RQ1–RQ4, and recommendations for investigators and developers.]

References

Appendix A

Ethical Approval Letter

[Insert CURES approval letter here once received – see guidance in the original CUPhesisTemplate2026.tex for the `includegraphics` command to place it.]

Appendix B

Chain of Custody and Experiment Log Templates

[To be completed.]